Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table? 2

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/1/0 0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129 2

Does R2 show up as an OSPF neighbor on R1? yes 2

Does R2 show up as an OSPF neighbor on R3? no 2

What does this information tell you?

The passive interface will stop to send Hello message. Consequently, the previous neighbor router will lose contact with this router until the interface is changed to "no passive" status. R2 lost contact with R1 and R3 initially when R2 set all interface to be passive. However, when the S0/1/0 is set to be "no passive", the connection between R2 and R1 restores. So, R3 can learn the 192.168.2.0/24 network through R1. 2

h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1
R2(config-router)# no passive-interface s0/1/1 2

i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/1/1 0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

2 65; Cost(R3 S0/1/1)+Cost(R2 G0/0)

Is R2 listed as an OSPF neighbor to R3? yes 2

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

To distinguish costs of different interfaces. The default OSPF reference bandwidth is 100 Mbps. In modern networks, many interfaces operate at speeds higher than 100 Mbps. With the default reference bandwidth, these high-speed interfaces would all have the same OSPF cost of 1. Changing the reference bandwidth allows OSPF to assign more accurate costs and better differentiate between high-speed links.

2

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

Cost = Reference bandwidth / Interface bandwidth

R1 to 192.168.3.0/24: R1-> R3-> LAN; Cost = Serial + GigabitEthernet = 100000 / 128 +1=782

R1 to 192.168.23.0/30: R1-> R3-> R2 (subnet); Cost= Serial + Serial = 100000 / 128 + 64 = 845

2

i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1562. Because all serial interfaces were configured with a bandwidth of 128 Kbps. The OSPF cost of each serial interface becomes approximately 781 (100000 / 128). R1 to 192.168.23.0/24:Cost= Serial + Serial = 100000 / 128 + 100000 / 128 = 1562

2

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

Because R1 will choose a path with the lowest cost (R1-> R2 -> R3 -> 192.168.3.0/24; cost= 781+781+1=1563), rather than the previous one (R1-> R3 -> LAN; cost= 1565+1=1566).

The cost on the S0/0/1 interface of R1 was set to 1565, increasing the total cost of the direct path from R1 to R3. As a result, the alternative path through R2 now has a lower total cost, so OSPF selects the route through R2.

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Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

   The router ID uniquely identifies each router in the OSPF network. It is used to establish and maintain OSPF neighbor relationships and to identify the source of routing updates. If router IDs are not controlled, duplicate or unexpected router IDs may occur, which can cause instability or incorrect OSPF operation.

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2. Why is the DR/BDR election process not a concern in this lab?

   Because the routers are connected using point-to-point serial links. DR and BDR elections only occur on multi-access networks such as Ethernet. However, R1, R2 and R3 all are in area 0.

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3. Why would you want to set an OSPF interface to passive?

   An interface is set to passive to prevent OSPF from sending hello packets and forming neighbor relationships on that interface while still advertising the network in OSPF. When a interface of a router connects to an end device, the router doesn't need to send any OSPF message through this interface. So, this interface could be set to passive, reducing traffic.

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4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

   The use of different areas. The network is divided into multiple areas connected to a backbone area (Area 0). This area structure reduces routing overhead and allows OSPF to scale to very large networks.

   4

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

   network 10.0.0.0 0.255.255.255 area 0

   4

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

   The EIGRP route. When multiple routing protocols provide routes to the same destination, the router selects the route with the lowest administrative distance rather than the value of their metrics. EIGRP has an administrative distance of 90, which is lower than RIP (120) and OSPF external (110).

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